THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{2}{5}$	Multiply numerators and multiply denominators, then simplify.
2	**(b)** 8	'Of' means multiply.
3	**(a)** $\frac{7}{3}$	Turn the fraction upside down.
4	**(c)** $\frac{1}{4}$	Dividing by n means multiplying by $\frac{1}{n}$.
5	**(c)** smaller than each of the fractions	Recall the idea from the chapter.
6	**(b)** $\frac{7}{3}$	Whole × denominator + numerator, over the denominator.
7	**(d)** $\frac{1}{3}$	Area = length × breadth.
8	**(a)** $\frac{3}{14}$	Multiply numerators and multiply denominators, then simplify.
9	**(a)** 12	'Of' means multiply.
10	**(a)** $\frac{2}{5}$	Turn the fraction upside down.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{3}{5} \times \frac{10}{21} = \frac{3\times10}{5\times21}; = \frac{30}{105}$ (1); Lowest terms: $\frac{2}{7}$ (1).	1+1
12	Reciprocal of $\frac{9}{16}$ is $\frac{16}{9}$; $\frac{3}{4} \times \frac{16}{9} = \frac{48}{36}$ (1); Answer $1\frac{1}{3}$ (1).	1+1
13	$\frac{3}{5} \times 40; = \frac{120}{5}$ (1); Answer 24 kg (1).	1+1
14	$\frac{4}{9} \times \frac{3}{8} = \frac{4\times3}{9\times8}; = \frac{12}{72}$ (1); Lowest terms: $\frac{1}{6}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Improper form: $2\frac{1}{3} = \frac{7}{3}$ (1); $\frac{7}{3} \times \frac{3}{5} = \frac{21}{15}$ (1); Answer $1\frac{2}{5}$ (1).	1+1+1
16	Pieces $= n \div \frac{3}{4}$ (1); $6 \times \frac{4}{3} = \frac{24}{3}$ (1); Pieces $= 8$ (1).	1+1+1
17	Area $= \frac{3}{4} \times \frac{2}{3}$; Area $= \frac{1}{2}$ m^2 (1); Perimeter $= 2(\frac{3}{4} + \frac{2}{3})$; $= 2 \times \frac{17}{12}$ (1); Perimeter $= 2\frac{5}{6}$ m (1).	1+1+1
	OR: Improper form: $1\frac{1}{2} = \frac{3}{2}$ (1); $\frac{3}{2} \times \frac{4}{9} = \frac{12}{18}$ (1); Answer $\frac{2}{3}$ (1).	1+1+1
18	Pieces $= n \div \frac{5}{6}$ (1); $10 \times \frac{6}{5} = \frac{60}{5}$ (1); Pieces $= 12$ (1).	1+1+1
	OR: Area $= \frac{5}{6} \times \frac{3}{5}$; Area $= \frac{1}{2}$ m^2 (1); Perimeter $= 2(\frac{5}{6} + \frac{3}{5})$; $= 2 \times \frac{43}{30}$ (1); Perimeter $= 2\frac{13}{15}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At first $\frac{3}{4} \times 120 = 90$ L (1); Used $\frac{2}{3} \times 90 = 60$ L (1); Left $= 90 - 60$ (1); $= 30$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 5 \div \frac{3}{4} = 6\frac{2}{3}$ (1); Full cakes $= 6$ (1); Flour used $= 6 \times \frac{3}{4} = 4\frac{1}{2}$ kg (1); Left over $= \frac{1}{2}$ kg (1); Final statement (1).	1+1+1+1+1
20	At first $\frac{5}{8} \times 80 = 50$ L (1); Used $\frac{4}{5} \times 50 = 40$ L (1); Left $= 50 - 40$ (1); $= 10$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 7 \div \frac{2}{3} = 10\frac{1}{2}$ (1); Full cakes $= 10$ (1); Flour used $= 10 \times \frac{2}{3} = 6\frac{2}{3}$ kg (1); Left over $= \frac{1}{3}$ kg (1); Final statement (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change ÷ to ×.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a + sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{3}{7}$	Dividing by n means multiplying by $\frac{1}{n}$.
2	**(a)** smaller	Recall the idea from the chapter.
3	**(c)** $\frac{17}{5}$	Whole $\times$ denominator + numerator, over the denominator.
4	**(d)** $\frac{3}{10}$	Area = length $\times$ breadth.
5	**(d)** $\frac{1}{2}$	Multiply numerators and multiply denominators, then simplify.
6	**(c)** 15	'Of' means multiply.
7	**(d)** $\frac{9}{8}$	Turn the fraction upside down.
8	**(c)** $\frac{1}{6}$	Dividing by n means multiplying by $\frac{1}{n}$.
9	**(c)** $\frac{1}{4}$	Recall the idea from the chapter.
10	**(d)** $\frac{7}{4}$	Whole $\times$ denominator + numerator, over the denominator.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reciprocal of $\frac{10}{3}$ is $\frac{3}{10}$; $\frac{5}{6} \times \frac{3}{10} = \frac{15}{60}$ (1); Answer $\frac{1}{4}$ (1).	1+1
12	$\frac{5}{8} \times 64$; $= \frac{320}{8}$ (1); Answer 40 litres (1).	1+1
13	$\frac{5}{12} \times \frac{6}{25} = \frac{5\times6}{12\times25}$; $= \frac{30}{300}$ (1); Lowest terms: $\frac{1}{10}$ (1).	1+1
14	Reciprocal of $\frac{21}{4}$ is $\frac{4}{21}$; $\frac{7}{8} \times \frac{4}{21} = \frac{28}{168}$ (1); Answer $\frac{1}{6}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Improper form: $3\frac{1}{4} = \frac{13}{4}$ (1); $\frac{13}{4} \times \frac{2}{13} = \frac{26}{52}$ (1); Answer $\frac{1}{2}$ (1).	1+1+1
16	Pieces $= n \div \frac{3}{8}$ (1); $9 \times \frac{8}{3} = \frac{72}{3}$ (1); Pieces $= 24$ (1).	1+1+1
17	Area $= \frac{2}{5} \times \frac{5}{8}$; Area $= \frac{1}{4}$ m^2 (1); Perimeter $= 2(\frac{2}{5} + \frac{5}{8})$; $= 2 \times \frac{41}{40}$ (1); Perimeter $= 2\frac{1}{20}$ m (1).	1+1+1
	OR: Improper form: $2\frac{2}{5} = \frac{12}{5}$ (1); $\frac{12}{5} \times \frac{5}{6} = \frac{60}{30}$ (1); Answer 2 (1).	1+1+1
18	Pieces $= n \div \frac{2}{3}$ (1); $12 \times \frac{3}{2} = \frac{36}{2}$ (1); Pieces $= 18$ (1).	1+1+1
	OR: Area $= \frac{7}{8} \times \frac{4}{7}$; Area $= \frac{1}{2}$ m^2 (1); Perimeter $= 2(\frac{7}{8} + \frac{4}{7})$; $= 2 \times \frac{81}{56}$ (1); Perimeter $= 2\frac{25}{28}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At first $\frac{2}{3} \times 90 = 60$ L (1); Used $\frac{3}{4} \times 60 = 45$ L (1); Left $= 60 - 45$ (1); $= 15$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 4 \div \frac{5}{6} = 4\frac{4}{5}$ (1); Full cakes $= 4$ (1); Flour used $= 4 \times \frac{5}{6} = 3\frac{1}{3}$ kg (1); Left over $= \frac{2}{3}$ kg (1); Final statement (1).	1+1+1+1+1
20	At first $\frac{4}{5} \times 150 = 120$ L (1); Used $\frac{5}{6} \times 120 = 100$ L (1); Left $= 120 - 100$ (1); $= 20$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 9 \div \frac{4}{5} = 11\frac{1}{4}$ (1); Full cakes $= 11$ (1); Flour used $= 11 \times \frac{4}{5} = 8\frac{4}{5}$ kg (1); Left over $= \frac{1}{5}$ kg (1); Final statement (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change $\div$ to $\times$.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a $+$ sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(c) $\frac{5}{14}$	Area = length × breadth.
2	(b) $\frac{1}{4}$	Multiply numerators and multiply denominators, then simplify.
3	(a) 9	'Of' means multiply.
4	(a) $\frac{11}{4}$	Turn the fraction upside down.
5	(c) $\frac{2}{9}$	Dividing by n means multiplying by $\frac{1}{n}$.
6	(c) 4	Recall the idea from the chapter.
7	(a) $\frac{25}{6}$	Whole × denominator + numerator, over the denominator.
8	(d) $\frac{1}{2}$	Area = length × breadth.
9	(d) $\frac{1}{6}$	Multiply numerators and multiply denominators, then simplify.
10	(d) 12	'Of' means multiply.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{2}{7} \times 49; = \frac{98}{7}$ (1); Answer 14 metres (1).	1+1
12	$\frac{7}{10} \times \frac{15}{28} = \frac{7\times15}{10\times28}; = \frac{105}{280}$ (1); Lowest terms: $\frac{3}{8}$ (1).	1+1
13	Reciprocal of $\frac{4}{3}$ is $\frac{3}{4}$; $\frac{2}{9} \times \frac{3}{4} = \frac{6}{36}$ (1); Answer $\frac{1}{6}$ (1).	1+1
14	$\frac{7}{12} \times 60; = \frac{420}{12}$ (1); Answer 35 rupees (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Improper form: $1\frac{2}{3} = \frac{5}{3}$ (1); $\frac{5}{3} \times \frac{3}{10} = \frac{15}{30}$ (1); Answer $\frac{1}{2}$ (1).	1+1+1
16	Pieces $= n \div \frac{7}{10}$ (1); $14 \times \frac{10}{7} = \frac{140}{7}$ (1); Pieces $= 20$ (1).	1+1+1
17	Area $= \frac{3}{10} \times \frac{5}{6}$; Area $= \frac{1}{4}$ m^2 (1); Perimeter $= 2(\frac{3}{10} + \frac{5}{6})$; $= 2 \times \frac{68}{60}$ (1); Perimeter $= 2\frac{4}{15}$ m (1).	1+1+1
	OR: Improper form: $4\frac{1}{2} = \frac{9}{2}$ (1); $\frac{9}{2} \times \frac{2}{3} = \frac{18}{6}$ (1); Answer 3 (1).	1+1+1
18	Pieces $= n \div \frac{5}{8}$ (1); $15 \times \frac{8}{5} = \frac{120}{5}$ (1); Pieces $= 24$ (1).	1+1+1
	OR: Area $= \frac{4}{9} \times \frac{3}{4}$; Area $= \frac{1}{3}$ m^2 (1); Perimeter $= 2(\frac{4}{9} + \frac{3}{4})$; $= 2 \times \frac{43}{36}$ (1); Perimeter $= 2\frac{7}{18}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At first $\frac{7}{9} \times 72 = 56$ L (1); Used $\frac{3}{7} \times 56 = 24$ L (1); Left $= 56 - 24$ (1); $= 32$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 6 \div \frac{7}{8} = 6\frac{6}{7}$ (1); Full cakes $= 6$ (1); Flour used $= 6 \times \frac{7}{8} = 5\frac{1}{4}$ kg (1); Left over $= \frac{3}{4}$ kg (1); Final statement (1).	1+1+1+1+1
20	At first $\frac{5}{6} \times 60 = 50$ L (1); Used $\frac{3}{5} \times 50 = 30$ L (1); Left $= 50 - 30$ (1); $= 20$ L (1); State all three answers (1).	1+1+1+1+1
	OR: Cakes $= 8 \div \frac{3}{10} = 26\frac{2}{3}$ (1); Full cakes $= 26$ (1); Flour used $= 26 \times \frac{3}{10} = 7\frac{4}{5}$ kg (1); Left over $= \frac{1}{5}$ kg (1); Final statement (1).	1+1+1+1+1

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